WORKED SOLUTION

STEP II 2023, Question 3

Pure • Polynomials and Derivatives

STEP II 2023, Question 3

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

The key idea

Differentiating q just shifts every term of the sum along by one, and the last derivative vanishes, so $q' = q - p$. That one relation drives everything. The three proofs that $q > 0$ then read it in three ways: as a fact about stationary points, as the derivative of $\mathrm{e}^{-x}q(x)$ (an integrating factor), and as the integral formula you get by integrating by parts again and again.

Solution

Throughout, write $p(x) = ax^n + \dots$ with $a \neq 0$. For $k \geq 1$, $p^{(k)}$ has degree less than n (or is zero), and $p^{(n+1)} \equiv 0$.

Part (i)

(a) A polynomial of odd degree tends to $+\infty$ at one end of the real line and to $-\infty$ at the other. Being continuous, it then takes the value 0 somewhere, by the intermediate value theorem. But $p(x) > 0$ for all x, so p has no real root, and n must be even.

Also $a > 0$: if $a < 0$ then $p(x) \to -\infty$ as $x \to \infty$, contradicting $p > 0$. In $q = p + p' + \cdots + p^{(n)}$ every term after the first has degree less than n, so q also has degree n with leading coefficient $a > 0$. Hence $q(x) \to +\infty$ as $x \to \infty$ (and, as n is even, also as $x \to -\infty$), so q certainly takes positive values. (If $n = 0$ then $q = p$ is a positive constant.)

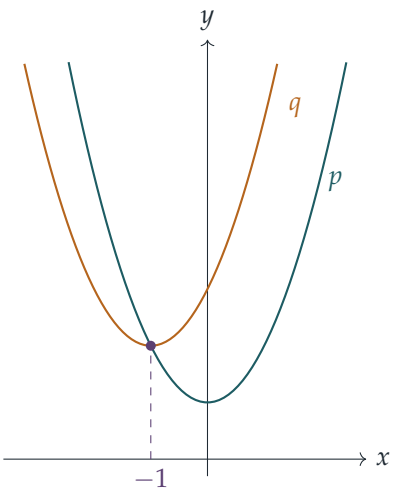
(b) Differentiating term by term,

$$q'(x) = \sum_{k=0}^{n} p^{(k+1)}(x) = \sum_{k=1}^{n+1} p^{(k)}(x) = \sum_{k=0}^{n} p^{(k)}(x) - p(x) + p^{(n+1)}(x) = q(x) - p(x),$$

since $p^{(n+1)} \equiv 0$ for a polynomial of degree n.

Part (ii)

(a) At a stationary point x_0 of $y = q(x)$ we have $q'(x_0) = 0$, so by (i)(b) $q(x_0) = p(x_0)$: the two curves meet there.



(The picture shows $p(x) = x^2 + 1$, for which $q(x) = x^2 + 2x + 3$. The minimum of q is at $x = -1$, exactly where the curves cross.)

Now for $q > 0$. If $n = 0$ then $q = p > 0$, so suppose $n \geq 2$. By (i)(a), $q(x) \to +\infty$ as $x \to \pm\infty$, so we can choose $R > 0$ with $q(x) > q(0)$ whenever $|x| > R$.

Ask yourself How do we know q has a lowest point at all, and that it is a stationary point?

On the closed interval $[-R, R]$ the continuous function q attains a minimum value at some x_0, and $q(x_0) \leq q(0)$. Outside the interval q is bigger than $q(0)$, so $q(x_0)$ is the minimum of q over the whole real line. Since x_0 is an interior point of $\mathbb{R}$ and q is differentiable, $q'(x_0) = 0$. By the first result, $q(x_0) = p(x_0) > 0$. So for every x,

$$q(x) \geq q(x_0) = p(x_0) > 0.$$

(b) Using (i)(b),

$$\frac{\mathrm{d}}{\mathrm{d}x}(\mathrm{e}^{-x}q(x)) = \mathrm{e}^{-x}(q'(x) - q(x)) = -\mathrm{e}^{-x}p(x) < 0$$

for all x, since $p > 0$. So $\mathrm{e}^{-x}q(x)$ is (strictly) decreasing.

For $q > 0$: since q is a polynomial, $\mathrm{e}^{-X}q(X) \to 0$ as $X \to \infty$ (exponentials beat powers). A decreasing function is never below its limit at $+\infty$, so $\mathrm{e}^{-X}q(X) \geq 0$ for all X. Then, for any x, strict decrease gives

$$\mathrm{e}^{-x}q(x) > \mathrm{e}^{-(x+1)}q(x+1) \geq 0,$$

and multiplying by $e^x > 0$ gives $q(x) > 0$.

(c) All the integrals here converge, because a polynomial in t times e^{-t} is integrable on $[0, \infty)$. Integrate by parts with $u = p(x+t)$ and $dv = e^{-t}\,dt$ (treating x as a constant):

$$\begin{aligned}\int_0^\infty p(x+t)e^{-t}\,dt &= \Big[-p(x+t)e^{-t}\Big]_0^\infty + \int_0^\infty p^{(1)}(x+t)e^{-t}\,dt \\ &= p(x) + \int_0^\infty p^{(1)}(x+t)e^{-t}\,dt,\end{aligned}$$

because $p(x+t)e^{-t} \to 0$ as $t \to \infty$. This is the first result.

Nothing in that calculation used anything about p except that it is a polynomial, so the same holds for every derivative: writing

$$I_k(x) = \int_0^\infty p^{(k)}(x+t)e^{-t}\,dt, \qquad \text{we have} \qquad I_k(x) = p^{(k)}(x) + I_{k+1}(x)$$

for $k = 0, 1, \ldots, n$. Also $I_{n+1}(x) = 0$, since $p^{(n+1)} \equiv 0$. Applying the relation repeatedly,

$$I_0(x) = p(x) + I_1(x) = p(x) + p^{(1)}(x) + I_2(x) = \cdots = \sum_{k=0}^{n} p^{(k)}(x) + I_{n+1}(x) = q(x),$$

which is the second result.

Ask yourself Why does this formula make $q(x) > 0$ obvious?

For each fixed x, the integrand $p(x+t)e^{-t}$ is continuous and strictly positive for every $t \geq 0$, because $p > 0$ everywhere. The integral of a continuous, strictly positive function over an interval of positive length is strictly positive. So $q(x) = I_0(x) > 0$ for all x.

Checks

- With $p(x) = x^2 + 1$ we get $q(x) = x^2 + 2x + 3 = (x+1)^2 + 2$. Indeed $q' = 2x + 2 = q - p$, the minimum of q is 2 at $x = -1$, and $p(-1) = 2$ as (ii)(a) predicts.
- For the same p at $x = 0$: $\int_0^\infty (t^2+1)e^{-t}\,dt = 2! + 1 = 3 = q(0)$.
- By computer, for 300 random everywhere-positive polynomials of degree up to 6, we confirmed that $q' = q - p$ exactly, that $q > 0$ and $e^{-x}q(x)$ decreases on a grid over $[-20, 20]$, and that numerical integration reproduces $q(x)$ to about ten significant figures.

Common slips

- In (ii)(a), assuming a minimum exists without a reason. It needs $q \to +\infty$ at both ends, which uses n even and $a > 0$ from (i)(a).
- In (ii)(b), concluding only $e^{-x}q(x) \geq 0$ from the limit. The strict decrease is what upgrades this to $q(x) > 0$.

- Forgetting that the boundary term $p(x+t)\mathrm{e}^{-t}$ must vanish as $t \to \infty$ before discarding it.

Beyond STEP

The function e^{-x} in (ii)(b) is an integrating factor, and (ii)(c) solves $q' - q = -p$ explicitly. The formula $\int_0^\infty p(x+t)\mathrm{e}^{-t}\,\mathrm{d}t$ is a Laplace-transform-style integral, and the identity $q = (1 + D + D^2 + \cdots)p = (1 - D)^{-1}p$, where D means differentiation, is a first taste of operator methods for linear differential equations. Boyce and DiPrima, *Elementary Differential Equations*, covers integrating factors and Laplace transforms; Abbott, *Understanding Analysis*, makes the extreme-value argument of (ii)(a) rigorous.

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